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A weighted twin support vector machine as a potential discriminant analysis tool and evaluation of its performance for near-infrared spectroscopic discrimination of the geographical origins of diverse agricultural products

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ABSTRACT

A weighted twin support vector machine (*w*TWSVM) was proposed as a potential discriminant analysis tool and its utility was evaluated for near-infrared (NIR) spectroscopic identification of the geographical origins of 12 different agricultural products including black soybean and garlic. In the *w*TWSVM, weights were applied on each variable in the sample spectra to highlight detailed NIR spectral features and the optimal weights to minimize the discrimination error were iteratively searched. Then, the weighted spectra were employed to determine the samples' geographical origins using a TWSVM adopting two non-parallel hyperplanes for the discrimination. For the performance evaluation, SVM, TWSVM, and *w*TWSVM were separately used for the two-group discriminations and their accuracies were comparatively analyzed. When the SVM and TWSVM accuracies were compared, the improvements by using the TWSVM were significant (95% confidence level) for 10 out of the 12 products. Moreover, the accuracy improvements with the *w*TWSVM against SVM were significant for all the 12 products. In the case of the TWSVM-*w*TWSVM accuracy comparison, the improvements by the *w*TWSVM were also significant for 10 products, thereby demonstrating superior discrimination performance of *w*TWSVM. Based on the overall results, the *w*TWSVM could be a potential chemometric tool for discriminant analysis and expandable to other areas such as spectroscopy-based biomedical disease diagnosis and forensic analysis.

1. Introduction

Vibrational spectroscopic methods such as Raman and near-infrared (NIR) spectroscopy have been widely used for the classification and discrimination of diverse samples in agricultural [1–6], pharmaceutical [7–13], and biomedical areas [8,14,15], since they provide relevant compositional information of a sample with fast and non-destructive spectral acquisition. Meanwhile, the chemical compositions of analyzed samples are generally complex. Therefore, subsequent sample spectral features are broad and indistinct due to extensive overlapping of individual component peaks and therefore, component specificity of the collected spectra is inferior. Under this circumstance, multivariate chemometric methods are routine choices for discriminant analysis to recognize minutely different spectral features among employed samples. For this purpose, linear discriminant analysis (LDA) [16–18], *k*-nearest neighbor (*k*-NN) [19,20], and support vector machine (SVM) [21,22] have been popularly utilized, where the SVM, effectively building hyperplane (boundary) between different sample groups, has become dominant owing to its superior discrimination performance.

To further improve the performance of the SVM, a twin SVM (TWSVM) was proposed [23]. Briefly, TWSVM builds two non-parallel hyperplanes simultaneously, first hyperplane closer to one group and second hyperplane as far as from the other, to maximize distance between two sample groups. Therefore, it potentially helps to identify classes more effectively. The detail mathematical description of TWSVM is included in the supplementary information. TWSVM has been readily applied for gesture classification based on surface electromyogram (sEMG) information of muscle activity from the skin surface [24,25], the

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Table 1
Total number of samples of each agricultural product including the numbers of imported and domestic samples with the corresponding percentages to the total.

	Product	Number of total samples	Number of imported samples	Number of domestic samples
1	Adlay	1074	526 (48.98%)	548 (51.02%)
2	Adzuki	1689	794 (47.01%)	895 (52.99%)
3	Black soybean	1628	619 (38.02%)	1009 (61.98%)
4	Garlic	1604	610 (38.03%)	994 (61.97%)
5	Millet	1238	569 (45.96%)	669 (54.04%)
6	Lingzhi	1029	484 (47.04%)	545 (52.96%)
7	Mung bean	1205	590 (48.96%)	615 (51.04%)
8	Omija	1302	664 (51.00%)	638 (49.00%)
9	Panicum	1445	780 (53.98%)	665 (46.02%)
10	Pine nut	740	377 (50.95%)	363 (49.05%)
11	Sesame seed	1722	827 (48.03%)	895 (51.97%)
12	White soybean	1748	804 (46.00%)	944 (54.00%)

identification of Alzheimer's disease at an early stage based on brain magnetic resonance image (MRI) data [26], and fault diagnosis for wireless sensors [27]. In the vibrational spectroscopy field, TWSVM was also employed for the NIR spectroscopic analysis of the hardness of licorice seeds and purity of maize seeds [28]. Except for this study, few relevant researches have been reported and therefore, an extended evaluation of the potential of the TWSVM in vibrational spectroscopic discriminant analysis is timely and versatile. Taking a step forward, an improved version of the TWSVM enabling enhanced discrimination performance would be beneficial.

With this motivation, a weighted-TWSVM (*w*TWSVM) was proposed in this study. As is known, the TWSVM estimates the hyperplanes based on an objective function that maximizing the distance between two groups in a classification, while the *w*TWSVM initially applies weights on each variable in the sample spectrum and searches optimal weighted spectra providing improved TWSVM accuracy by setting a loss function as an objective function. The loss function is equivalent to a mean squared error (MSE), which is the squared difference between the reference and predicted values by the TWSVM. The procedure of searching optimal weights on spectra is repeated until the loss function (equivalently MSE) is minimized. Thus, the *w*TWSVM is expected to be

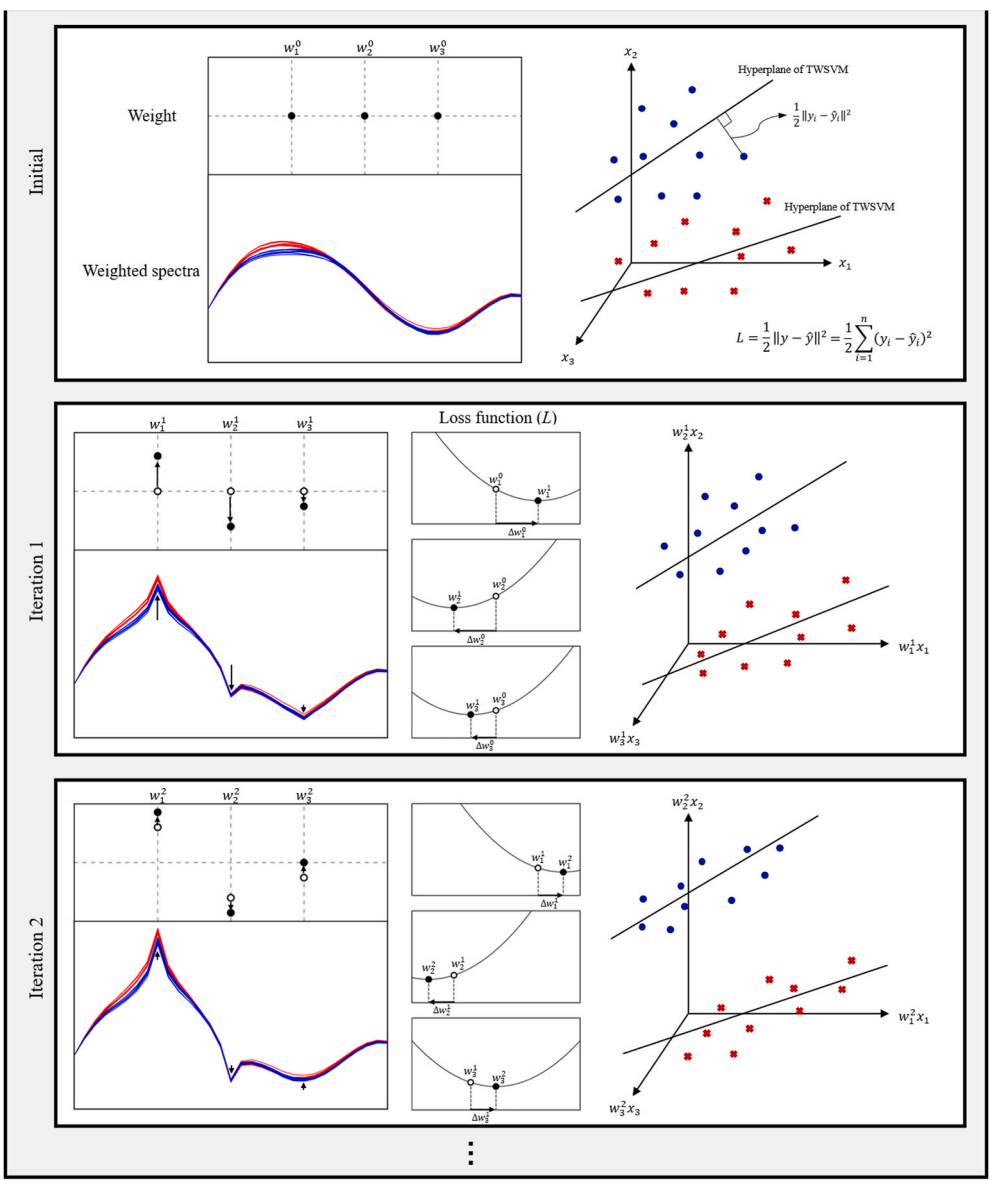

Fig. 1. Graphical description of the *w*TWSVM procedure including weight updating in the sample spectra.

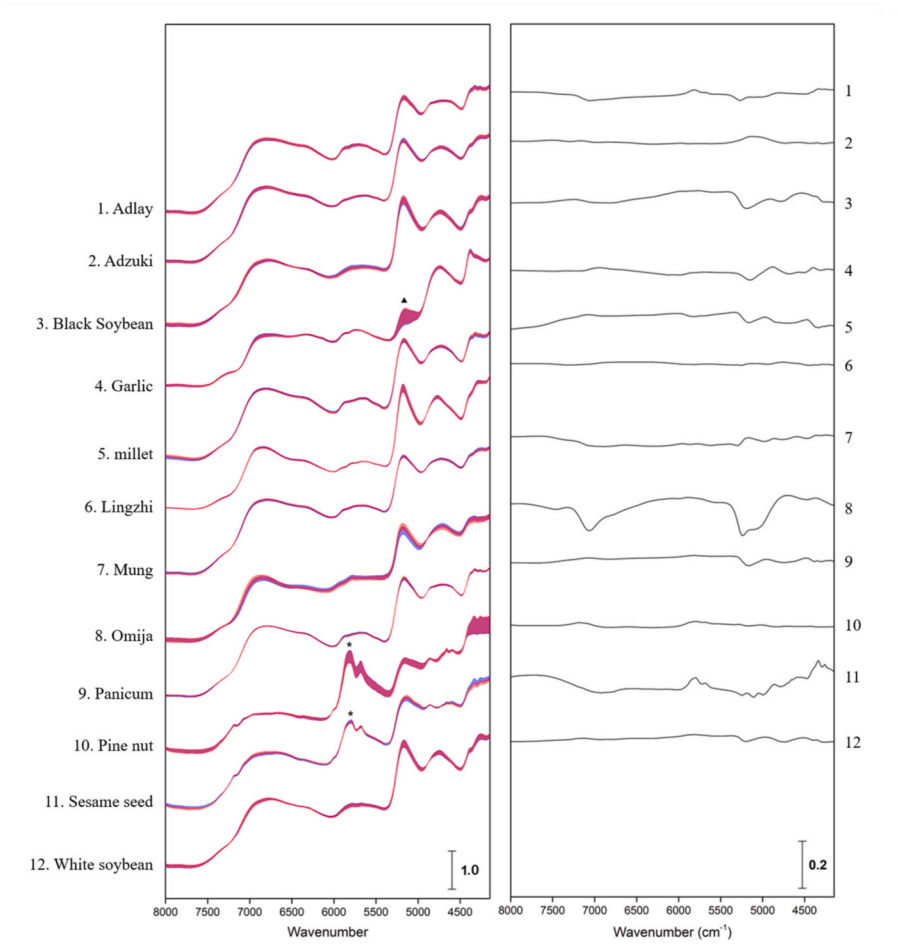

Fig. 2. Average NIR spectra of the imported (red) and domestic (blue) samples of the 12 different agricultural products (left plot). The shades superimposed with the average spectra correspond to the standard deviation at each wavenumber, showing the degree of spectral variation among the samples. The difference spectra for each product obtained by subtracting the average imported sample spectrum from domestic spectrum are also shown (right plot). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

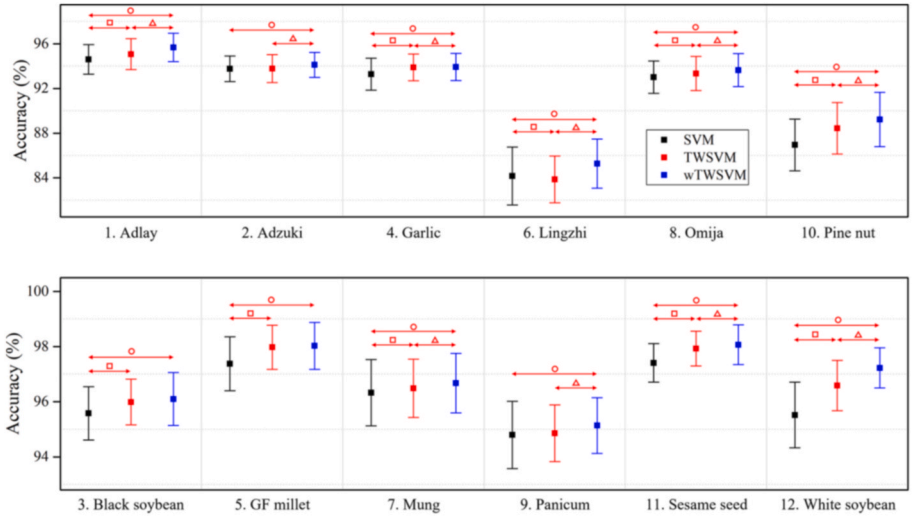

Fig. 3. Accuracies and corresponding standard deviations resulting from the discrimination of samples according to their geographical origins in each agricultural product using the SVM, TWSVM, and *w*TWSVM. The cases yielding a *p*-value less than 0.05, where the difference is significant with a 95% confidence level, are designated by the red arrows. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

effective in improving the TWSVM-based discrimination accuracy. The detailed mathematical and graphical descriptions are presented in the following section. Meanwhile, SVM combining with weighting techniques were also demonstrated, for example, to improve outlier sensitivity problem by considering weights according to the relative importance of data points to hyperplane estimation [29], to enhance classification performance by applying different weights to each class to solve the class imbalance problem [30], and to construct optimal margin of hyperplane using moving weighted average to generate both weighted and unweighted support vectors [31].

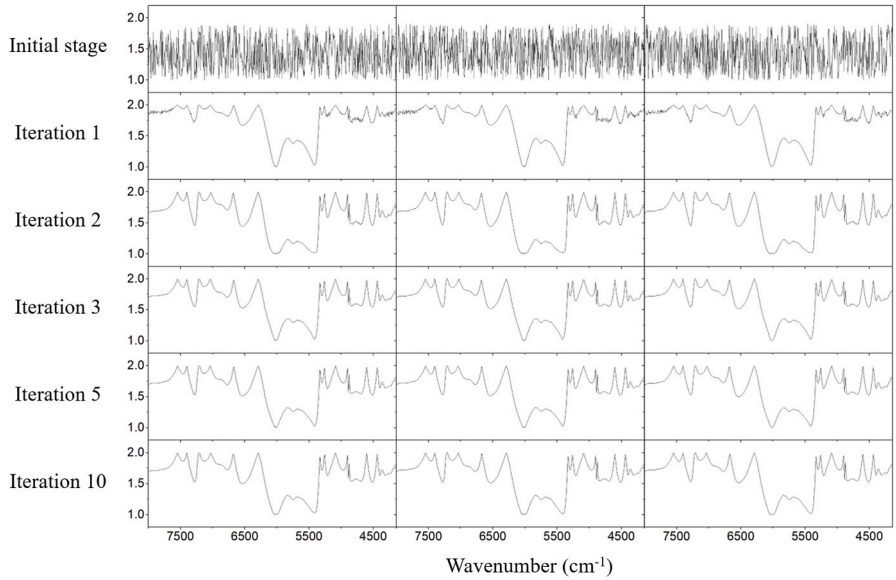

Fig. 4. Initial random weights across the whole range (first row) and the updated weights after 1, 2, 3, 5, and 10 iterations (second through sixth row). The plots in each column show the weight update process when the initial random weights are different.

For the evaluation of the *w*TWSVM performance, NIR spectroscopic discriminations of the geographical origins of 12 different agricultural products (1: adlay, 2: adzuki, 3: black soybean, 4: garlic, 5: millet, 6: lingzhi 7: mung bean, 8: omija, 9: panicum, 10: pine nut, 11: sesame seed, 12: white soybean) were attempted. To ensure reliability of the evaluation, 12 different cases were studied and a large number of samples (740–1748 samples per each agricultural product) collected over 4–6 years was employed in each case to represent diverse cultivation areas with a wide compositional variation of samples. Since the compositions of adopted agricultural products are complex and subsequent NIR spectral features are expected to be indistinct and overlapped, highlight of more detailed sample discriminant information by the application of proper weights on the sample spectra would be advantageous to enhance the discrimination ability. Using each NIR dataset of the 12 agricultural products (Table 1), SVM, TWSVM, and *w*TWSVM were separately used to discriminate the samples according to their geographical origins and the resulting discrimination accuracies were comparatively analyzed. This analysis simultaneously allows comparison of the discrimination performances of the SVM and TWSVM.

2. Experimental section

2.1. NIR spectral datasets of 12 different agricultural products

All of the NIR sample spectra of the 12 different agricultural products (1: adlay, 2: adzuki, 3: black soybean, 4: garlic, 5: millet, 6: lingzhi 7: mung bean, 8: omija, 9: panicum, 10: pine nut, 11: sesame seed, 12: white soybean) employed in this study were kindly provided from the National Agricultural Products Quality Management Service (NAQS, Gimcheon, Korea), an official government organization certifying the geographical origins of agricultural products. The samples cultivated in diverse regions were cautiously collected over 4–6 years by the NAQS to avoid duplicate sampling and to simultaneously include a wide compositional variation in the dataset. Table 1 shows the total number of samples for each agricultural product, including the numbers of imported and domestic samples with the corresponding percentages to the total. The numbers of imported and domestic samples in each dataset were fairly well balanced, while the percentages of domestic black soybean and garlic samples were approximately 62%. Among the imported mung bean samples, 70% and 30% of them were originated from China and Myanmar, respectively. In the case of imported sesame seed, 40%, 30%, and 30% of the samples were from China, India, and Sudan, respectively. The imported white soybean set was composed of 80% Chinese and 20% American samples. For the remained 9 products except the 3 products mentioned above, all the imported samples were originated from China. All the domestic samples were collected in diverse regions in Korea.

For NIR spectral acquisition, garlic and lingzhi samples were freeze and hot-air dried, respectively. Then, these dried garlic and lingzhi samples were powdered and the subsequent powders (20-mesh) were measured. For omija, the hot-air dried samples were directly measured without sample grinding. These pretreatments were accomplished right after the sample collection and diffuse reflectance NIR spectra the pretreated samples were immediately collected with respect to a reference ceramic standard. For the remained 9 agricultural products, NIR spectra of raw (naturally dried) samples in common markets without further pretreatments were measured instantly after the sample collection. Each NIR spectrum had 552 variables (absorbance values) over the spectral range from 8000 to 4140 cm^{-1}. Multiplicative scatter correction (MSC) of the raw spectra was conducted using MATLAB version 2017a (The Math Works Inc., MA, USA).

2.2. Schematic and brief mathematical description of the wTWSVM

Fig. 1 shows the graphical description of the procedure of the *w*TWSVM including the weight updating in sample spectra. The top plot shows the initial stage where random weights ($w_1{}^0$, $w_2{}^0$, $w_3{}^0$) are applied at three different wavenumbers (or variables) in the sample spectra. Based on the weighted spectra, two hyperplanes are constructed using the TWSVM for the discrimination of two sample groups and a loss function (L) is obtained. In the next stage (Iteration 1, middle plot), new weights ($w_1{}^1$, $w_2{}^1$, $w_3{}^1$) able to minimize the discrimination error using the L of the TWSVM are determined.

$$L = \frac{1}{2}\|y - \hat{y}\|^2 = \frac{1}{2}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2$$

Here, y is the reference value and $\hat{y}$ is the predicted value which is a linear combination of weight w. Therefore, L can be expressed as a quadratic equation for weights and finding optimal weights (w^*) is then carried out by utilizing the partial differential for weights.

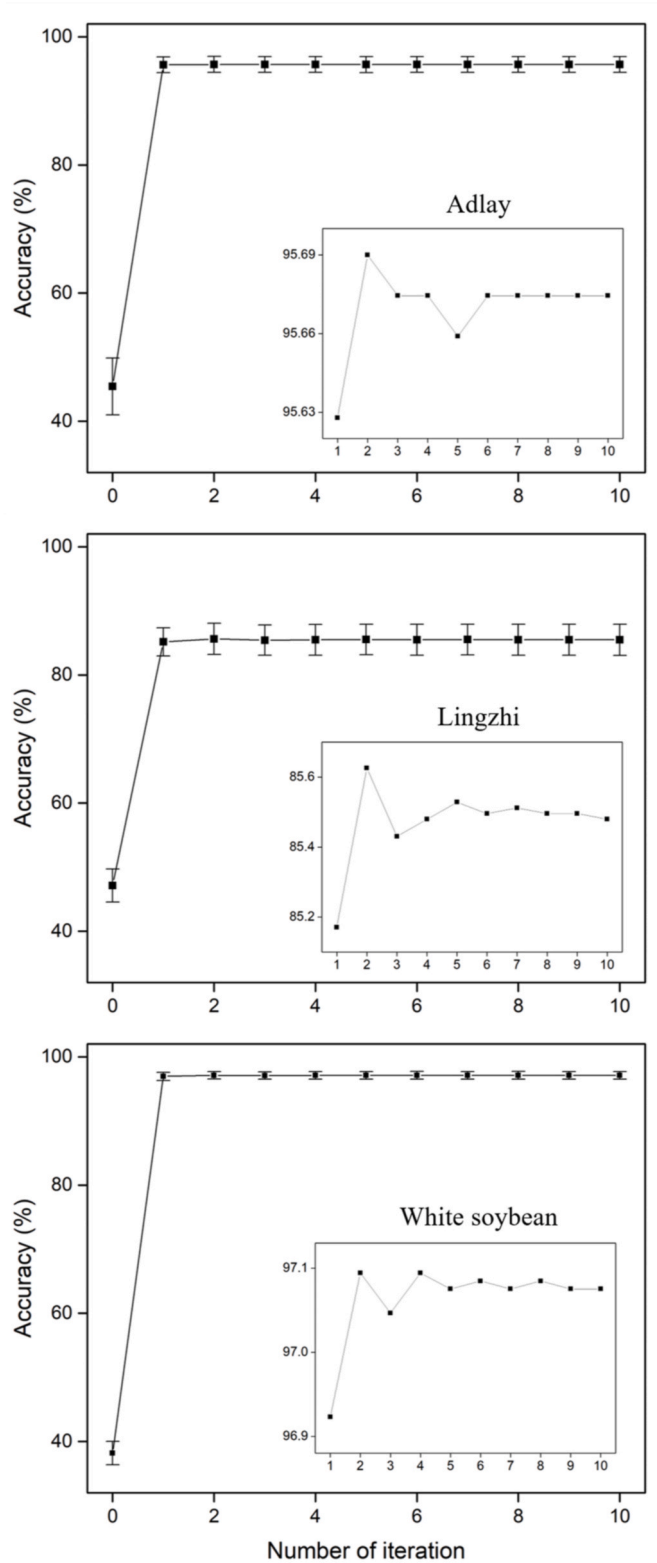

Fig. 5. Variations of the accuracy with weights updated by up to 10 iterations in the discrimination of adlay (top), lingzhi (middle), and white soybean (bottom) samples. The inset in each plot magnifies the accuracies at the 1st through 10th iteration.

$$w^* = w + \eta\Delta w = w + 2\eta w \sum_{i=1}^{n}(y_i - \widehat{y}_i)x_i \sum_{i=1}^{n}\alpha_i y_i x_i$$

Here, η is the learning rate and Δw is the gradient of the weight. Therefore, during iteration 1, the spectra weighted by the updated weights ($w_1{}^1$, $w_2{}^1$, $w_3{}^1$) are used for the TWSVM. The bottom plot shows iteration 2 to find the next updated weights ($w_1{}^2$, $w_2{}^2$, $w_3{}^2$) to minimize L further. This iteration step is repeated until the variation of the weight converges. More detailed mathematical description of the procedure of the *w*TWSVM is included in the supplementary information.

2.3. Discriminant analysis using the SVM, TWSVM, and wTWSVM

It is necessary to determine the optimal parameters in the SVM, TWSVM, and *w*TWSVM for discriminant analysis. For a kernel function, both linear and radial basis kernels were used for the evaluation. The cost constant for the two kernel functions was chosen by performing a grid search varying from 0.1, 1, 10, to 100. For each dataset, 80% of the samples were randomly selected to train a model and the remaining 20% of the samples were used to validate the developed model. This procedure was repeated 1000 times for cross-validation. The SVM, TWSVM, and *w*TWSVM were executed using MATLAB version 2020a and R package. The relevant MATLAB and R codes of *w*TWSVM are provided in the supplementary information.

3. Results and discussions

3.1. NIR spectral features of 12 different agricultural products

Fig. 2 shows the average NIR spectra of the imported (red) and domestic (blue) samples of the 12 different agricultural products (left plot). The MSC-treated spectra of the samples are shown in the figure. The shades superimposed with each average spectrum indicate the corresponding standard deviations at each wavenumber, providing the degree of spectral variation among the samples in a group. As can be seen, the average spectral features of the 12 agricultural products are different owing to their dissimilar sample compositions. Meanwhile, the spectral shapes of pine nut (#10) and sesame seed (#11) are considerably different from those of the other products, including the high intensity peaks in the 6000-5500 cm^{-1} range (marked by the asterisks) due to the high content of edible oil in these products. In the case of garlic (#4), the intensity of the peak at 5200 cm^{-1} (marked by the triangle) is considerably smaller than the rest.

The right plot in Fig. 2 presents the difference spectra for each product obtained by subtracting the average imported sample spectrum from the average domestic spectrum. The spectral differences are quite large in the cases of omija (#8) and sesame seed (#11), suggesting that the difference in average component compositions between the domestic and imported samples is relatively large, while they are small in the cases of lingzhi (#6) and pine nut (#10). The large magnitude of the spectral difference does not always guarantee better two-group separation since the degree of spectral variation among the samples in a group (the shade around the average spectrum) also governs the accuracy of discrimination. Overall, for all of the agricultural products, the spectra of imported and domestic samples of each product are similar to each other without noticeable differences in certain ranges. Therefore, sensible recognition of minute spectral differences between the two groups is necessary for accurate discrimination.

3.2. Comparison of the discrimination accuracies using the SVM, TWSVM, and wTWSVM

Using the NIR sample spectra of each agricultural product, two-group discrimination was performed separately using the SVM, TWSVM, and *w*TWSVM, and the obtained discrimination accuracies were compared. Fig. 3 shows the accuracies obtained from the

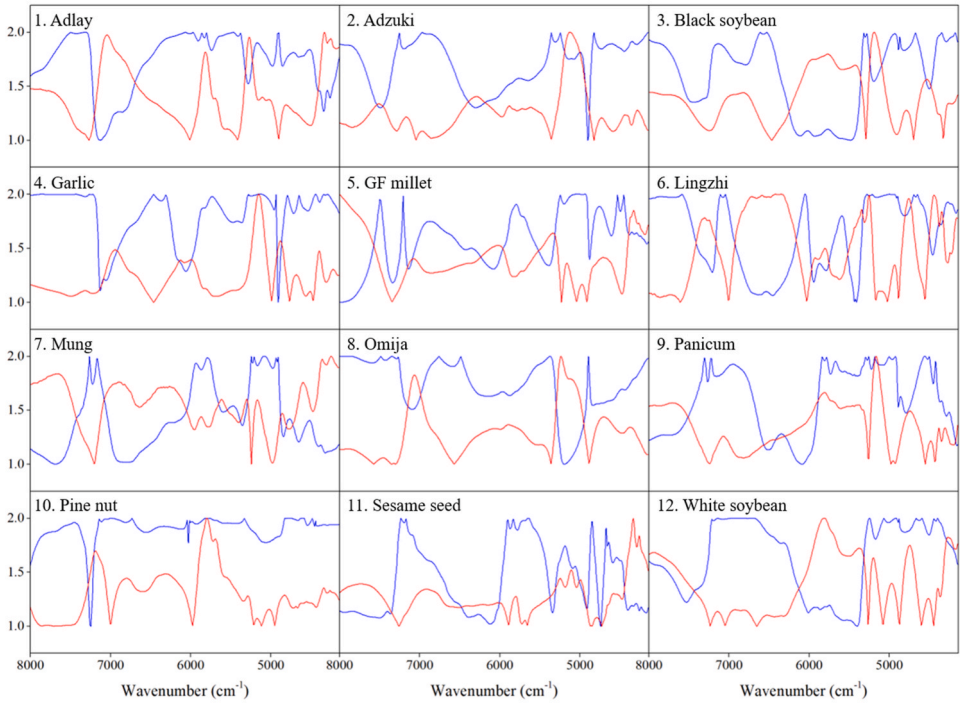

Fig. 6. Weights updated by 10 iterations over the 8000-4140 cm^{-1} range (blue) employed in the *w*TWSVM for the discrimination of samples' geographical origins of each agricultural product and scaled difference spectrum (red) between the imported and domestic sample spectra. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

discrimination of samples according to their geographical origins of each agricultural product. The corresponding standard deviations (the vertical bars) of accuracies in the 1000-time cross validations are also shown. A linear kernel was employed for all 3 cases. The employed cost constant in the SVM was 1. The two cost constants necessary to execute the TWSVM and *w*TWSVM were 10 and 10 for each case. The real values of accuracies, and corresponding recalls and precisions are presented (Table S1) in the supplementary information. The recalls and precisions were calculated by assigning the domestic sample as true. The agricultural products yielding discrimination accuracies in the 84–96% (6 products) and 94–100% ranges (6 products) are separately presented in the top and bottom plots, respectively. For each product, the use of the *w*TWSVM (blue) yields a higher discrimination accuracy compared to those using the SVM (black) and TWSVM (red). In fact, accuracy improvement by the *w*TWSVM was observed in all 12 cases.

To assess the significance of accuracy improvement on a statistical basis, the ANOVA test was initially performed whether the difference among the accuracies using the three methods was significant or not. The ANOVA test showed that the differences were significant with a 95% confidence level (*p*-values less than 0.05) for all the 12 products. Next, Duncan's multiple range test using studentized range distribution was chosen for pair-wise comparison [32,33] in three respective cases: SVM vs. TWSVM, SVM vs. *w*TWSVM, and TWSVM vs. *w*TWSVM, designated by square, circle, and triangle, respectively, in Fig. 3. Cases yielding a *p*-value less than 0.05 are designated by red arrows in the figure. When a *p*-value is larger than 0.05, no red arrow is present for the corresponding case. When the SVM and TWSVM accuracies are compared, the improvements by the TWSVM are significant for 10 out of the 12 products (except adzuki (#2) and panicum (#9)) as designated by the arrows with squares. This demonstrates that the TWSVM is more effective for the discrimination of the samples' geographical origins compared to the SVM, while the improvements are not significant for the remaining 2 products.

The comparison of the SVM and *w*TWSVM accuracies reveals that the improvements by the *w*TWSVM are significant for all the 12 products (indicated by the arrows with a circle). In the case of the TWSVM-*w*TWSVM accuracy comparison, the improvements by the *w*TWSVM are also significant for 10 products (the arrows with triangles). This observation clearly demonstrates that the discrimination performance of the *w*TWSVM is superior to those of the SVM and TWSVM. The overall comparison confirms that the *w*TWSVM is effective and versatile to improve the discrimination accuracy.

The discriminations of imported and domestic samples in each agricultural product are feasible since the NIR spectral features of the samples in both groups are different. The different spectral features are attributed to dissimilar compositions of the imported and domestic samples originated from diverse factors such as the genetic origin, metabolic pathway, and cultivation conditions as previously reported [34–36]. The thorough elucidation of botanical justification supporting the minute spectral differences in each agricultural product in Fig. 2 needs extensive phytochemical investigations; therefore, it would be beyond the scope of this study, which intends to demonstrate the utility of *w*TWSVM for discriminant analysis using the broadly overlapping NIR spectra of the agricultural products.

3.3. Investigation of the detailed characteristics of the *w*TWSVM

Visualization of the weight update process is helpful to better understand the characteristics of the *w*TWSVM. For this purpose, the weight update performed on the white soybean spectra are shown in Fig. 4 (the plot in the left column) as an example. At the initial stage, random weights in the range from 1 to 2 are applied over the whole range, as shown in the top plot. Therefore, the weight spectrum looks like random noise. With the 1st weight update (Iteration 1), the overall shape of the weights over the range becomes completely altered. The new weights on each wavenumber yielding a minimal discrimination error at this stage were determined as described earlier. Meanwhile, the small noise patterns still remain at the left and right regions. With the

2nd update (Iteration 2), the weight shape changes a little and the previous noise patterns in the 1st update disappear. With the 3rd, 5th, to 10th updates, the corresponding weight shapes vary without noticeable differences. The plots at the middle and right columns show the same weight update process when the initial random weights are different from that in the left plot. Although the initial weights randomly vary in each case, the shapes of the updated weights among the three cases are nearly analogous. This confirms that the final weights found by the iteration ultimately become the same, regardless of the initial random weights.

Fig. 5 shows the variations of the accuracy with the increase of the weight update iteration up to 10 times in the discrimination of adlay (#1), lingzhi (#6), and white soybean (#12) samples. The inset in each plot magnifies the accuracies at the 1st through 10th iteration. The zero iteration corresponds to the application of random weights at the initial stage. The accuracies at the zero iteration are around or below 50% due to the application of the random weight, which is surely not descriptive of the samples' geographical origins. Even with just one weight update (Iteration 1), the accuracies substantially improve in all cases and minutely fluctuate during the 2nd-5th weight updates. Then, the accuracies become steady in the 6th-10th weight updates. Therefore, the discrimination results of the *w*TWSVM shown in Fig. 3 (Table S1) were obtained using the spectra updated by 10 iterations.

Fig. 6 shows the weights updated 10 times over the 8000-4140 cm^{-1} range (blue) employed in the *w*TWSVM for the discriminant analysis of the 12 agricultural products. For comparison, the scaled difference spectrum (red) is also displayed together in each plot. The difference spectrum was acquired by subtracting the average imported sample spectrum from the domestic sample spectrum and then dividing by the difference between the minimal and maximal values for scaling. In the case of adlay (#1), the variation of weights across the spectral range is somewhat opposite to the shape of the scaled difference spectrum. This demonstrates that, in the *w*TWSVM, higher weights are imposed on the regions where the magnitude of the spectral difference between the domestic and imported samples is smaller, for example, the 7400–6900 and 5000-4800 cm^{-1} ranges. Analogous trends were observed for the remaining 11 agricultural products. This observation suggests that the *w*TWSVM tends to highlight the minutely dissimilar spectral features by spectral weighting with simultaneous minimization of the discrimination error.

4. Conclusion

Compared to a widely adopted SVM, the TWSVM improved the accuracies in the NIR spectroscopic discrimination of samples' geographical origins for 10 out of 12 products. Moreover, the accuracy improvements were significant for all 12 products by using the *w*TWSVM. In the case of TWSVM-*w*TWSVM accuracy comparison, the improvements by the *w*TWSVM are also significant for 10 products. Therefore, the *w*TWSVM searching optimal weights on the spectra with minimizing the discrimination error could be a potential chemometric tool for spectroscopic data-based discriminant analyses such as biomedical disease discrimination and forensic sample identification. Other than the NIR spectral data, different molecular spectroscopic information such as infrared (IR) and Raman spectra will be employed to further evaluate the performance of the *w*TWSVM. The development of an evolved *w*TWSVM effectively dealing with classification of more than 2 groups is underway in parallel.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.talanta.2021.122973.

Credit author statement

Daeil Jang: Software, Validation, Formal analysis, Investigation, Data curation, Writing – review & editing, Visualization **Woosuk Sohng:** Software, Validation, Formal analysis, Investigation, Data curation, Writing – review & editing, Visualization **Kyungjoon Cha:** Conceptualization, Methodology, Resources, Writing – original draft, Writing – review & editing, Supervision **Hoeil Chung:** Conceptualization, Methodology, Resources, Writing – original draft, Writing – review & editing, Supervision.

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